Aakash Kulkarni

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PROFESSIONAL SUMMARY

Cloud-focused Computer Science graduate with hands-on AWS experience and AWS Certified Solutions Architect Associate (SAA-C03) credential. Built and deployed distributed systems on AWS infrastructure, working with EC2, S3, RDS, Lambda, and VPC configurations while automating deployments through infrastructure as code practices.

Strong foundation in networking fundamentals including TCP/IP, DNS, routing, and security policies. Configured and managed network architectures for distributed applications, implemented security groups and access controls, and troubleshot connectivity issues across multi-service environments during research and development projects.

Experienced with Infrastructure as Code using Terraform and Docker, automating infrastructure deployments and maintaining consistent environments across development and production. Wrote Python and Bash scripts to orchestrate complex workflows on distributed systems, demonstrating ability to automate repetitive tasks and improve operational efficiency.

Collaborative team player with proven ability to work cross-functionally with platform, application, and research teams. Communicated technical concepts clearly to diverse stakeholders, participated in Agile workflows, and contributed to team success through knowledge sharing and documentation of architectures and runbooks.

Eager to learn and adapt to new technologies in fast-moving environments. Stayed current with AWS services and best practices through continuous learning, successfully applied emerging tools to solve real-world problems, and demonstrated willingness to take on new challenges in cloud networking and security domains.

TECHNICAL SKILLS

Cloud & Networking:	AWS (VPC, EC2, S3, RDS, Lambda, Route 53, Security Groups, NACLs, IAM), Cloud Infrastructure, TCP/IP, DNS, Routing, Firewalls, Load Balancing, Hybrid Network Architectures
Infrastructure as Code:	Terraform, Docker, Infrastructure Automation, Configuration Management, CI/CD Pipelines
Programming & Scripting:	Python, Bash, JavaScript, Java, C++, SQL, Node.js
Monitoring & Troubleshooting:	Log Analysis, Performance Monitoring, Connectivity Troubleshooting, System Debugging, Network Diagnostics
Security:	AWS Security Groups, Network ACLs, IAM Policies, Access Control, Network Segmentation, Security Best Practices
Databases:	MySQL, PostgreSQL, MongoDB, Aurora, Database Performance Optimization
Development Tools:	Git, GitHub, Linux, Postman, JIRA, REST APIs
Frameworks & Platforms:	React, Spring Boot, Django, Flask, Express, Microservices Architecture
Methodologies:	Agile/Scrum, Documentation, Code Reviews, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant Mar 2025 - Jun 2025

College of Engineering — Oregon State University *Corvallis, Oregon*

- Mentored over 100 students in Parallel Programming, explaining concepts like multi-threading, memory management, synchronization, and distributed systems.
- Held weekly office hours to help students with assignments and projects related to high-performance computing and parallel systems design.
- Graded programming projects and provided feedback to help students improve their coding skills, system design thinking, and understanding of distributed computing.

Graduate Research Assistant Apr 2024 - Mar 2025

ANSWER Lab, Oregon State University — Advisor: Dr. Manish Motwani *Corvallis, Oregon*

- Working on HeteroBugDetect, a framework that uses large language models to automatically find bugs in scientific applications that only appear when running on specific hardware platforms like GPUs versus CPUs.
- Built the testing infrastructure using AFL++ fuzzer, configured it to work across different computing platforms, and wrote Python scripts to automate the entire testing process on HPC clusters.
- Developed grammar-based input generation system that creates valid test inputs for LAMMPS, a molecular dynamics simulator. The grammar ensures inputs are syntactically correct while exploring different execution paths.
- Set up and configured LAMMPS on both CPU-only and heterogeneous systems with GPUs, then ran differential testing to compare outputs and identify platform-specific bugs.
- Analyzed test results to verify bug reports, distinguishing between real bugs and false positives. Successfully identified 8 known bugs and discovered 2 previously unknown bugs in LAMMPS.

- Deployed and managed distributed testing infrastructure across HPC clusters, troubleshooting connectivity issues between nodes and ensuring reliable job orchestration across diverse computing environments.
- Automated infrastructure setup and configuration using Python and Bash scripts, reducing manual deployment time and ensuring consistent environments across multiple testing platforms.
- Documented testing methodologies, system architectures, and troubleshooting procedures in detailed runbooks, making it easier for team members to understand and maintain the testing infrastructure.
- Co-authored research paper accepted at IEEE AIxSE 2025. Contributed to writing methodology sections, participated in research discussions, and helped revise the paper based on reviewer feedback.
- Learned how to design experiments for systems research, manage large-scale testing on HPC infrastructure, and communicate technical findings to the research community.

Associate Software Developer Jun 2021 - Mar 2023

Zemoso Technologies *Hyderabad, India*

- Worked on TruValidate Console at TransUnion, a platform that unified fraud detection, risk analysis, and credit reporting into a single onboarding experience for customers across global markets.
- Implemented micro-frontend architecture using Single SPA, which allowed different teams to develop and deploy their frontend modules independently without waiting for other teams to finish their work.
- Designed and built GraphQL APIs to replace some of the existing REST endpoints, reducing the amount of unnecessary data being sent over the network and making the onboarding pages load faster for users.
- Built RESTful APIs using Spring Boot and Java that handled credit and fraud-related requests. These services were part of a microservices architecture that needed to be reliable and handle high traffic volumes.
- Troubleshot production issues by digging through logs, profiling JVM performance, and identifying bottlenecks in the code. Fixed bugs and optimized slow queries to improve platform stability.
- Collaborated with frontend developers, backend engineers, and platform teams to integrate the different microservices and microfrontends, making sure everything worked together smoothly for the end-to-end onboarding flow.
- Worked on a second project building full-stack applications with React, Node.js, Django, and MySQL. Developed responsive frontend components and backend services with REST APIs.
- Set up Django REST Framework backends with authentication, database models, and API endpoints. Tested APIs using Postman and worked with PostgreSQL and MongoDB for data storage.
- Participated in Agile sprints with daily stand-ups, sprint planning, and retrospectives. Reviewed code from teammates and used Git for version control across the team.
- Learned new technologies on the job and adapted to different tech stacks across projects, picking up Spring Boot, GraphQL, Single SPA, and Django REST Framework as needed.

Mobile Application Developer Intern Jul 2021 - Sep 2021

FindMind Analytics *Remote*

- Developed ZuPay, a mobile banking app for teenagers with parental controls, using Flutter and JavaScript for cross-platform deployment on iOS and Android.
- Integrated Firebase backend services for authentication, real-time database, and cloud storage, which handled user data and enabled real-time synchronization across devices.
- Launched the app on both iOS and Android app stores. The app reached over 10,000 active users and received positive feedback from parents and teenagers using the platform.
- Wrote clean code following mobile development best practices, participated in code reviews with the team, and used Git for version control throughout the development process.

EDUCATION

Oregon State University Mar 2023 – Jun 2025

Master's in Computer Science — GPA: 3.91/4.0 *Corvallis, Oregon*

Advisor: Dr. Manish Motwani — **Relevant Courses:** High Performance Computing, Deep Learning, Software Engineering Methodologies, Parallel Programming, Mobile Application Development

Jawaharlal Nehru Technological University Jul 2018 – Aug 2022

Bachelor's in Computer Science — GPA: 7.25/10.0 *Hyderabad, India*

Relevant Courses: Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Networks, Distributed Systems

SELECTED PROJECTS

AWS Infrastructure Automation — *Terraform, AWS (VPC, EC2, RDS, S3), Python, Bash* Jan 2025

- Built automated infrastructure deployment for a web application using Terraform to provision VPCs, subnets, security groups, EC2 instances, RDS databases, and S3 buckets on AWS.
- Configured VPC with public and private subnets across multiple availability zones, set up routing tables, and implemented network ACLs and security groups to control traffic flow and secure resources.
- Wrote Terraform modules for reusable infrastructure components, making it easy to deploy consistent environments for development, staging, and production with different configurations.
- Automated DNS management using Route 53, setting up hosted zones and routing policies to direct traffic to the application load balancers.
- Created Python scripts to monitor infrastructure health, check connectivity between services, and send alerts when issues were detected.
- Documented the infrastructure architecture and wrote runbooks for common operational tasks like scaling instances, updating security groups, and troubleshooting connectivity issues.

Multi-Region Cloud Deployment — *Docker, AWS (EC2, ELB, CloudWatch), Bash, Python* Nov 2024

- Deployed containerized application across multiple AWS regions using Docker and EC2, with load balancers distributing traffic between regions for high availability.
- Set up CloudWatch monitoring to track application performance, network latency, and error rates across regions. Configured alarms to notify the team when metrics exceeded thresholds.
- Implemented security groups and IAM roles to control access between services, following the principle of least privilege to minimize security risks.
- Wrote Bash scripts to automate deployment processes, including building Docker images, pushing to registries, and updating EC2 instances with new container versions.
- Troubleshot cross-region connectivity issues by analyzing VPC peering configurations, route tables, and security group rules to ensure services could communicate properly.

Django E-Commerce Platform — *Django, PostgreSQL, Docker, AWS (RDS, EC2), Nginx* Sep 2024

- Built full-stack e-commerce platform using Django with product catalog, shopping cart, order management, and payment processing features.
- Designed RESTful APIs with JWT authentication for secure user management and session handling.
- Deployed application on AWS using EC2 instances behind an Application Load Balancer, with RDS PostgreSQL database for data storage.
- Containerized the application using Docker and wrote deployment scripts to automate the process of updating the production environment.
- Configured Nginx as a reverse proxy to handle SSL termination and serve static files, improving application performance and security.
- Optimized database queries and added indexes to improve page load times for product searches and catalog browsing.

RESEARCH PUBLICATIONS

Manish Motwani, **Aakash Kulkarni**, Yunhan Qiao, Matthew Davis, and Ziyan Chen. "LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications." In *Proceedings of the IEEE International Conference on AI x Software Engineering (AIxSE)*, 2025.

CERTIFICATIONS

AWS Certified Solutions Architect Associate (SAA-C03) - January 2026

AWS Cloud Certification - AICTE and Edu Skills Foundation - January 2022

AWS Cloud Essential Skills - Coursera - June 2021

ACHIEVEMENTS

1st Place - OSU Winter 2024 Hackathon (March 2024): Developed "Guitar Hub," a web application with guitar tuner and AI-powered song analysis tools using OpenAI and Hugging Face models, delivering a working product under tight hackathon deadlines.